

NFC Book IV L4 Equivalence Certificate

Schema v0.40.0

L4 Spine Release Candidate

Completes the full branch-to-source dependency chain

Generated by NFC Governance Engine v40

Contents

Book IV L4 Certificate

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Book IV completes the chain. **U** exists (Thm. IV.5.1), is unique (by factorization role), and every lawful realization factors through it. The full spine dependency chain from any branch endpoint back to the source is now machine-traceable in the governance engine. This is an L4 spine-pilot shell — the canonical Book IV is the authoritative source.

Book IV L4 Equivalence Certificate

L4 gate: PASSED (10/10 criteria)

Universal Completion (Thm. IV.5.1): [U]

Source Uniqueness: [U]

Descendants Factor Through U: [U]

Full chain machine-traceable: True

Completes the full spine dependency chain. With Book IV records, every branch endpoint datum $E = F(U)$ now has a machine-traceable path from E back through $\text{cor:IV.descendants-factor} \rightarrow \text{thm:IV.universal-completion} \rightarrow \text{def:IV}$.

L4 Criteria

Check	St.	Observed
l4.iv.universal-completion-U	✓	thm:IV.universal-completion status = U
l4.iv.source-uniqueness-U	✓	thm:IV.source-uniqueness status = U
l4.iv.descendants-factor-U	✓	cor:IV.descendants-factor status = U
l4.iv.ORA-rungs-complete	✓	all 5 rungs [U]
l4.iv.governing-U	✓	thm:IV.governing status = U
l4.iv.POT-definitions	✓	POT-object, POT-category, candidate-U all present
l4.iv.unified-emergence-conditional	✓	status = C, constraints = ['K0-7', 'Phase-A', 'Books-I-
l4.iv.full-chain-traceable	✓	IV: True, BookI: True, BookV: True, BookVII: True

Check	St.	Observed
l4.iv.prose-completeness	✓	100.0% (19/19)
l4.iv.xref-annotations	✓	19/19